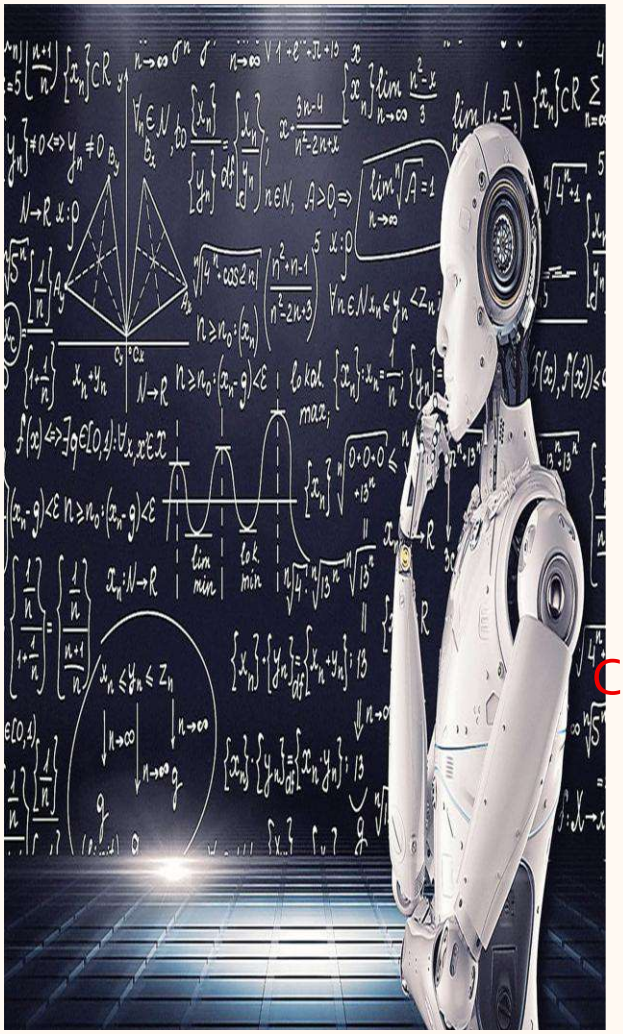
AI- BASED SMART MESS FOOD WASTE &DEMAND PREDICTION SYSTEM



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ABSTRACT

The Smart Mess Food Waste and Demand Prediction System is an Artificial Intelligence-based solution designed to minimize food wastage in hostel mess environments. The system uses machine learning techniques to predict the number of students expected to dine on a particular day based on factors such as previous attendance, day of the week, and special events. Based on the predicted demand, the system suggests the optimal quantity of food to be prepared. Additionally, rule-based logic is applied to generate recommendations for reducing over-preparation or shortages. This project demonstrates the practical application of AI and Machine Learning in solving real-world problems related to resource optimization and sustainability.

INTRODUCTION

Food wastage is a significant issue in institutional settings such as college hostels and mess facilities. Due to the lack of proper planning and demand estimation, large quantities of food are often either wasted or insufficient to meet student needs. Traditional methods rely on rough estimations, which are inefficient and inaccurate.

With the advancement of Artificial Intelligence and Machine Learning, it is now possible to analyze past data and predict future outcomes more effectively. This project introduces a smart system that leverages data-driven techniques to estimate daily attendance in a mess and recommend the appropriate amount of food preparation.

The system combines machine learning for prediction and rule-based logic for decision-making, making it an intelligent and practical solution. It not only improves efficiency but also contributes to sustainability by reducing food wastage.

PROBLEM DEFINATION

In hostel mess systems, food preparation is generally based on assumptions rather than accurate data analysis. This leads to two major problems:

1. Over-preparation of food, resulting in significant wastage
2. Under-preparation, causing inconvenience to students

There is a need for a system that can:

- Accurately predict the number of students who will consume food.
- Optimize food preparation accordingly.
- Provide actionable recommendations to reduce waste.

OBJECTIVES

The main objectives of this project are:

- To develop a system that predicts daily student attendance in the mess.
- To estimate the quantity of food required based on predicted attendance.
- To reduce food wastage and improve resource utilization.
- To implement machine learning techniques for real-life problem solving.
- To provide intelligent recommendations using rule-based logic.

REPORT ANALYSIS

- FUNCTIONAL REQUIREMENTS

THE SYSTEM SHOULD ACCEPT INPUTS SUCH AS:
DAY OF THE WEEK
PREVIOUS ATTENDANCE
EVENT INFORMATION
IT SHOULD PREDICT THE EXPECTED NUMBER OF STUDENTS
IT SHOULD CALCULATE THE REQUIRED FOOD QUANTITY
IT SHOULD PROVIDE RECOMMENDATIONS BASED ON PREDICTION

- NON-FUNCTIONAL REQUIREMENTS

THE SYSTEM SHOULD BE SIMPLE AND USER-FRIENDLY
IT SHOULD PROVIDE QUICK RESULTS
IT SHOULD BE EFFICIENT AND LIGHTWEIGHT
IT SHOULD BE EASILY SCALABLE FOR FUTURE IMPROVEMENTS

- HARDWARE REQUIREMENTS

BASIC COMPUTER/LAPTOP
MINIMUM 4GB RAM

- SOFTWARE REQUIREMENTS

PYTHON (3.X)
LIBRARIES:
PANDAS

NUMPY
SCIKIT-LEARN

FEASIBILITY ANALYSIS

1. **Technical Feasibility**: Easily implementable using basic ML libraries.
2. **Economic Feasibility**: Low cost, no special hardware required.
3. **Operational Feasibility**: Can be used in real hostel/mess environments.